Strand 2 · Computations and Algebraic Relationships

This is the **largest strand** on the test (about 15 of 40 questions). It is all about **lines, relationships, and equations.** Master it and you master most of the Math test.

Standards covered:

- ○ **8.4A** — Similar right triangles show slope is the same between any two points on a line.
- ★ **8.4B** — Graph proportional relationships; interpret the unit rate as the slope.
- ★ **8.4C** — Use a table or graph to find the rate of change (slope) and the y-intercept.
- ○ **8.5A** — Linear proportional situations as y = kx. ○ **8.5B** — Non-proportional as y = mx + b (b ≠ 0).
- ○ **8.5E** — Direct variation. ○ **8.5F** — Distinguish proportional from non-proportional. ○ **8.5H** — Examples of each.
- ★ **8.5G** — Identify functions from ordered pairs, tables, mappings, graphs.
- ★ **8.5I** — Write an equation y = mx + b to model a linear relationship.
- ○ **8.8A** — Write one-variable equations/inequalities with variables on both sides. ○ **8.8B** — Write a real-world problem for a given equation. ★ **8.8C** — Model and solve them.
- ○ **8.9A** — Identify a system's solution from the intersection of two graphs.

Part A · Slope and Rate of Change (8.4A ○, 8.4B ★, 8.4C ★)

Slope measures how steep a line is — how much *y* changes for each change in *x*:

$$\textbf{slope} = m = (\text{change in } y) \div (\text{change in } x) = (y_2 - y_1) / (x_2 - x_1)$$

- **8.4A — Why slope is constant.** Right triangles drawn to any two points on the same line are **similar**, so rise ÷ run is the **same everywhere** — that ratio is the slope.
- **8.4B — Proportional graphs.** A proportional relationship passes through the **origin (0, 0)**; equation **y = kx**; the **unit rate is the slope k.**
- **8.4C — From a table or graph.** Slope = (change in y) ÷ (change in x); the **y-intercept** is the y-value when **x = 0.**

Worked example (8.4C). Table: (0, 2), (1, 5), (2, 8). Slope = (5 − 2) ÷ (1 − 0) = **3**; y-intercept = **2**; equation y = 3x + 2.

Watch Out: Rise over run — count the **vertical** change first, then horizontal.

Try It. A line through (0, 0) and (4, 10) — slope? *(10 ÷ 4 = 2.5.)*

Part B · Proportional vs. Non-Proportional (8.5A/B/E/F/H ○)

	Proportional	Non-Proportional

Equation	y = kx	y = mx + b (b ≠ 0)
Graph	Line **through the origin**	Line **not** through the origin
Table	Constant ratio $y/x = k$	y/x is **not** constant

- **8.5E — Direct variation** is a proportional relationship y = kx. If y = 15 when x = 3, then k = 5, so y = 5x.
- **8.5F / 8.5H — Telling them apart.** Does the line pass through (0, 0)? Yes → proportional. A starting amount (y-intercept ≠ 0) → non-proportional.

Worked example (8.5B). A gym charges a $20 sign-up fee plus $5 per visit: **y = 5x + 20.** The $20 fee (b ≠ 0) makes it **non-proportional.**

Watch Out: A per-item rate with no starting fee is proportional; add any flat fee and it becomes non-proportional.

Try It. Is y = 7x proportional? *(Yes — passes through the origin, b = 0.)*

Part C · Functions (8.5G ★)

A **function** pairs **each input (x) with exactly one output (y).** In tables/mappings, an x-value may not repeat with two different y-values. On graphs, use the **vertical line test** — if any vertical line crosses the graph more than once, it is **not** a function.

Worked example. Is {(1, 3), (2, 5), (1, 7)} a function? **No** — input 1 pairs with both 3 and 7.

Try It. Does a non-vertical straight line pass the vertical line test? *(Yes — it is a function.)*

Part D · Writing Linear Equations (8.5I ★)

To write **y = mx + b**, find the **slope m** and the **y-intercept b.**

Worked example. A plant is 4 cm tall and grows 2 cm each week. Slope m = **2**; y-intercept b = **4**; equation **y = 2x + 4.**

Strategy: "Starting amount" is almost always **b**; "rate per unit" is almost always **m.**

Try It. A taxi charges $3 to start plus $2 per mile. Equation for cost y after x miles? *(y = 2x + 3.)*

Part E · One-Variable Equations and Inequalities (8.8A/B ○, 8.8C ★)

When the variable is on **both sides,** collect variable terms on one side and numbers on the other.

Worked example (8.8C). Solve 3x + 5 = x − 7. Subtract x → 2x + 5 = −7 Subtract 5 → 2x = −12 Divide by 2 → **x = −6.**

8.8A write such equations from a situation; **8.8B** do the reverse — describe a real situation the equation could model.

Watch Out: Whatever you do to one side, do to the **other** side — keep it balanced.

Try It. Solve 4x − 3 = 2x + 9. *(2x = 12, x = 6.)*

Part F · Systems of Two Linear Equations (8.9A ○)

Two lines graphed together **intersect** at the point that **satisfies both equations.** That point **(x, y)** is the **solution** of the system.

Worked example. y = x + 1 and y = −x + 5 cross at **(2, 3).** Check: 3 = 2 + 1 ✓ and 3 = −2 + 5 ✓.

Try It. If two lines cross at (−1, 4), what is the solution? *(x = −1, y = 4.)*

Strand 2 Checklist

- ☐ 8.4A–C Find and interpret slope, unit rate, and y-intercept
- ☐ 8.5A/B/E/F/H Tell proportional from non-proportional; use y = kx and y = mx + b
- ☐ 8.5G Identify functions (including the vertical line test)
- ☐ 8.5I Write an equation for a linear relationship
- ☐ 8.8A–C Write and solve one-variable equations with variables on both sides
- ☐ 8.9A Identify a system's solution from the intersection of two graphs